LMP Equations & Notation

Series shorthand — quick reference for Manuscripts 1–19

Particle and phase scales

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}, \qquad E^2 = p^2c^2 + m^2c^4, \qquad K = (\gamma - 1)mc^2.$$

$$\nu_C = \frac{mc^2}{h}, \qquad \nu_E = \frac{E}{h} = \gamma\nu_C, \qquad h = 2\pi\hbar.$$

Energy divided by h is a frequency-equivalent scale; a proper-time phase identity alone does not make a particle an operational clock.

Clock and lapse notation

t, τ t is the chosen reference coordinate time; τ is proper time on the clock's worldline. State the reference before comparing rates.

α Static lapse: $\mathrm{d}\tau = \alpha\,\mathrm{d}t$. In Schwarzschild exterior, $\alpha(r) = \sqrt{1 - 2GM/(rc^2)}$.

$\delta\tau$ Inverse rate, $\delta\tau \equiv \mathrm{d}t/\mathrm{d}\tau$. For a static clock, $\delta\tau_{\mathrm{static}} = 1/\alpha$. It is dimensionless, not an elapsed-time interval.

N Alternative generic lapse notation in the Tolman manuscript. With the same time normalization, $N \equiv \alpha$ and $1/N = 1/\alpha = \delta\tau_{\mathrm{static}}$.

ϵ_{grav} Static inverse-rate excess: $\epsilon_{\mathrm{grav}} \equiv \delta\tau_{\mathrm{static}} - 1$. For moving clocks, $\delta\tau = \mathrm{d}t/\mathrm{d}\tau$ also includes motion.

Mass symbols Use M for a gravitating source when distinguished from a test particle; use m for the particle. Some earlier expansions use M for particle mass.

Weak field and gradients

Define the static gravitational clock potential by $\psi \equiv \ln\delta\tau_{\mathrm{static}} = -\ln\alpha$. For a static metric with zero shift, let v_{loc} be a clock's speed relative to local static observers and $\gamma_{\mathrm{loc}} \equiv (1 - v_{\mathrm{loc}}^2/c^2)^{-1/2}$. Then

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \frac{\alpha}{\gamma_{\mathrm{loc}}}, \qquad \delta\tau = \frac{\gamma_{\mathrm{loc}}}{\alpha}, \qquad \ln\delta\tau = \psi + \ln\gamma_{\mathrm{loc}}.$$

For $|\Phi|/c^2 \ll 1$ and $v_{\mathrm{loc}}^2/c^2 \ll 1$, the first-order moving-clock ledger is

$$\delta\tau - 1 \simeq -\frac{\Phi}{c^2} + \frac{v_{\mathrm{loc}}^2}{2c^2}.$$

For an infinity-normalized reference, take $\Phi \to 0$ at infinity. The approximation $\psi \simeq -\Phi/c^2$ describes the static gravitational clock potential; the full moving-clock inverse rate also includes the local-motion term.

Define ∇_{proper} per unit proper spatial distance and take $\mathbf{g}_{\mathrm{in}}$ inward:

$$\mathbf{g}_{\mathrm{in}} = c^2\nabla_{\mathrm{proper}}\ln\delta\tau_{\mathrm{static}}, \qquad \mathbf{a}_{\mathrm{hold}} = -\mathbf{g}_{\mathrm{in}}.$$

These are static-observer relations; moving-clock rates require a specified trajectory and frame.

Mass, light-time, and Schuler scales

$$\tau_g \equiv \frac{GM}{c^3}, \qquad t_R \equiv \frac{R}{c}, \qquad x_R \equiv \frac{GM}{Rc^2} = \frac{\tau_g}{t_R}.$$

For a static Schwarzschild clock at R, $\delta\tau_{\mathrm{static}} = (1 - 2x_R)^{-1/2}$ exactly; $\epsilon_{\mathrm{grav}} \simeq x_R$ only to first order.

f_g $f_g \equiv g_{\mathrm{N}}/c$: acceleration-derived inverse-time scale, not an oscillation frequency. For an ideal sphere, $g_{\mathrm{N}} = GM/R^2$.

$$\omega_{\mathrm{S}}^2 = \frac{g_{\mathrm{N}}}{R}, \qquad f_{\mathrm{S}} = \frac{\omega_{\mathrm{S}}}{2\pi}, \qquad 4\pi^2 f_{\mathrm{S}}^2 t_R = f_g, \quad 4\pi^2 f_{\mathrm{S}}^2 t_R^2 = x_R.$$

This Schuler bridge is algebraic in the ideal spherical-Earth model, not causal. R/c converts length to time; it is not a signal path through Earth.

Thermal and horizon scales

For static thermal equilibrium in a static spacetime (Tolman relation),

$$NT = T_*, \qquad \frac{T}{T_*} = \frac{1}{N} = \delta\tau_{\mathrm{static}}.$$

Here T is local temperature and T_* is the chosen reference normalization; this is not a general law for arbitrary motion or nonequilibrium states.

$$T_{\mathrm{U}} = \frac{\hbar a}{2\pi c k_{\mathrm{B}}}, \qquad f_{\mathrm{U}} \equiv \frac{k_{\mathrm{B}}T_{\mathrm{U}}}{h} = \frac{a}{4\pi^2 c}.$$

f_{U} is thermal energy expressed in frequency units, not a spectral line or oscillation.

For a four-dimensional Schwarzschild black hole under standard semiclassical thermodynamics,

$$T_H = \frac{\hbar}{8\pi k_{\mathrm{B}}\tau_g}, \qquad S_{\mathrm{BH}} = \frac{k_{\mathrm{B}}A}{4\ell_P^2}, \qquad 2T_H S_{\mathrm{BH}} = Mc^2.$$

Area ratio and entropy in bits:

$$n_A \equiv \frac{A}{\ell_P^2}, \qquad \frac{S_{\mathrm{BH}}}{k_{\mathrm{B}}} = \frac{n_A}{4}, \qquad n_{\mathrm{bit}} = \frac{n_A}{4\ln 2}.$$

These are area units and entropy units, not literal horizon tiles or stored bits.

Planck and action shorthand

$$\ell_P^2 = \frac{\hbar G}{c^3}, \qquad t_P^2 = \frac{\hbar G}{c^5}, \qquad M\ell_P^2 = \hbar\tau_g.$$

For reduced Compton time $\widetilde{\tau}_C \equiv \hbar/(Mc^2)$,

$$\tau_g\widetilde{\tau}_C = t_P^2.$$

In Manuscript 18, $\omega_g \equiv 1/\tau_g$ is an inverse-time scale, not a physical angular velocity; $I_* \equiv M\ell_P^2$ is a defined inertia scale, not a body's measured moment of inertia.

Kerr shorthand

For an uncharged Kerr black hole,

$$\chi \equiv \frac{cJ}{GM^2}, \qquad \sigma \equiv \sqrt{1 - \chi^2}, \qquad |\chi| \le 1,$$

$$r_\pm = \frac{GM}{c^2}(1 \pm \sigma), \qquad \frac{r_\pm}{c} = \tau_g(1 \pm \sigma).$$

The last quantities are Boyer–Lindquist coordinate radii divided by c, not proper times read at the horizon.

Reading rules

f **vs.** ω $f = \omega/(2\pi)$ for an actual cyclic/angular frequency pair. A defined inverse-time scale need not represent periodic motion.

$\simeq$ Approximation, with order and regime stated; do not promote first-order weak-field results to exact identities.

Model labels Say whether a result is an exact algebraic identity, a Newtonian/weak-field model, a semiclassical input, a prediction, or a measurement.